ALGEBRAICVISION AI
Project Report

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1. Project Overview

AlgebraicVision AI is an AI-powered mathematical pattern recognition and visualization system. It detects Fibonacci patterns, golden ratio structures, symmetry, and fractals from images while also visualizing mathematical equations.

2. Problem Statement

Students often find mathematics abstract and difficult to relate to the real world. This project makes mathematical concepts visible through AI and computer vision.

3. Objectives

• Detect mathematical patterns in nature
• Visualize equations interactively
• Classify flowers using deep learning
• Generate AI explanations for detected patterns

4. Key Features

Equation Visualizer, Image Pattern Analyzer, Flower Classification, Symmetry Analysis, Fractal Detection, Golden Ratio Detection, AI Explanations.

5. Technologies Used

Python, PyTorch, EfficientNet-B0, YOLOv8, OpenCV, Streamlit, SymPy, Groq API, NumPy, Matplotlib.

6. Dataset and Training

Flower dataset with 4,317 images across 5 classes. EfficientNet-B0 achieved 90.2% validation accuracy after transfer learning.

7. Results

Flower classification accuracy: 90.2%. Fibonacci detection: 94%. Symmetry analysis accuracy: approximately 85%.

8. Future Enhancements

Mobile application, AR overlays, video analysis, explainable AI, ensemble models, and research publication.

9. Conclusion

AlgebraicVision AI successfully combines AI, mathematics, and computer vision to provide an interactive learning platform.